50-DAY DATA ANALYST CHALLENGE

EXCEL · DAY 01

Understanding and Exploring Business Data

Domain	Retail & E-commerce
Role	Junior Data Analyst
Estimated time	60–90 minutes
Difficulty	Beginner
Submission	Completed workbook + README on GitHub

1. Business situation

You have joined an online retail company as a Junior Data Analyst. The sales manager has shared a transaction-level sales file and wants you to explore the data before the weekly review. Your job is to understand the structure, organize the data, calculate basic measures, and communicate what you discover.

2. Files provided

Day01_Raw_Sales_Data.xlsx — contains the Raw_Sales worksheet and a Data_Dictionary worksheet. Treat the raw worksheet as source data; do not overwrite or delete the original records.

3. Your tasks

A. Understand the data

• Identify the number of records and fields.

• Classify each field as text, date, whole number, currency/decimal, or percentage.

• Explain what one row represents and what each row's Order_ID is used for.

B. Organize the workbook

• Keep the original data unchanged. Create a separate worksheet named Analysis.

• Convert the sales range into an Excel Table and give the table a meaningful name.

• Apply suitable date, currency, percentage, and number formats.

• Use sorting and filtering to inspect the records.

C. Calculate business measures

• Add calculated fields for Gross Sales, Discount Amount, and Net Sales on your Analysis worksheet.

• Calculate total units sold, total gross sales, total discount amount, and total net sales.

• Calculate average net sales per order, and identify the highest and lowest net-sales orders.

• Use appropriate Excel formulas. Do not type calculated answers manually.

D. Investigate the business

• Which region has the largest total net sales?

• Which product has the largest total net sales value?

• Which category contributes more to net sales?

• What do you notice about the discounts and their effect on sales?

• Write two evidence-based observations and one recommendation for the sales manager.

4. Business thinking – explain your reasoning

Do not submit numbers alone. For each conclusion, state the evidence from your analysis. If you think the company should take an action, explain why. Also mention one additional piece of information you would request before making a final business recommendation.

5. Submission requirements

Required file	What it must contain
Day01_Analysis.xlsx	Original Raw_Sales preserved; separate Analysis worksheet; formulas; clear forma ting; Business_Insig
README.md	Challenge name, short approach, key findings in your own words, and any assump ions or difficul ies.

6. GitHub folder structure

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50-Day-Data-Analyst-Challenge/
■■■ Excel/
■■■ Day-01/
■■■ Day01_Analysis.xlsx
■■■ README.md
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7. Submission checklist

■ Original raw data preserved ■ Analysis worksheet completed ■ Formulas used
■ Business_Insights worksheet completed ■ README.md written ■ GitHub link ready

Important: Work independently. Do not copy another student's workbook or submit someone else's analysis. Your reasoning and explanation matter as much as the numerical result.